

Mitigation of High-Frequency Common-Mode Voltage in 2-Level 3-Phase Voltage Source Inverters

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Abstract—This paper introduces a novel method to mitigate the high-frequency common-mode voltage (CMV) issue in 2-level 3-phase (2L-3Ph) voltage-source inverters (VSIs). Up to now, several solutions have been proposed to mitigate CMV, such as using passive/active filters, introducing additional switches, and utilizing specially designed pulse-width modulation (PWM) methods. Since the other solutions increase the cost and complexity, specially designed PWM solutions have become attractive and more promising. However, existing literature generally states that PWM solutions primarily reduce the high-frequency switching components of CMV, though CMV due to dv/dt remains because of switching instants. This study proposes a novel PWM method to eliminate CMV from dv/dt by synchronizing the modulation patterns of the legs of the VSI. This synchronization method is infeasible for direct utilization in conventional sinusoidal PWM (SPWM); therefore, this study introduces a third-harmonic square-wave injection and an adaptive carrier phase shift method to make it possible. In an ideal case, the proposed method completely removes the generated dv/dt , but, in practice, a small impact of dv/dt might still be observed because of the effects of the non-idealities of the switches and gate driving circuits. The proposed method is demonstrated through an in-depth analysis of analytical results. Finally, the effectiveness and feasibility of the proposed method were validated using an experimental setup with SiC devices.

Index Terms—Common-mode voltage, motor drive, pulse width modulation synchronization, sinusoidal pulse width modulation, voltage source inverter.

I. INTRODUCTION

Wide-band gap (WBG) devices based on silicon carbide (SiC) and gallium nitride (GaN) have become game changers in motor drive systems owing to their superior properties, which provide numerous benefits, such as greater compactness and higher efficiency [1]–[3]. However, WBG devices must be fast-switched to fully utilize these benefits, and this results in high dv/dt at switching instances, which generates a common mode voltage (CMV) [4]–[7]. This CMV might result in a bearing current and a shaft voltage, leading to premature bearing breakdown, which is one of the most frequent failure types in motors [8]–[10]. Additionally, the CMV may also cause stress on the motor windings. Therefore, researchers and industry professionals have investigated various CMV mitigation methods to prevent such failures, which are classified into three main groups: connection-side, motor-side, and inverter-side solutions [11], [12].

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First, connection-side solutions involve adding filters between the inverter and motor, such as common-mode chokes or shortened paths using resonant structures [13]–[16]. These methods effectively attenuate the common-mode current through the bearing or parasitic capacitances of the motor. However, the passive elements utilized can be bulky and costly. Second, motor-side solutions use new insulation materials in bearings and windings or grounded shaft [6], [7], [17]–[19]. Such approaches directly mitigate bearing currents or avoid breaking the winding insulation; however, their effectiveness is prone to motor design. Moreover, new types of materials may have unfavorable mechanical properties and may be expensive. Lastly, inverter-side solutions involve the implementation of new topologies or the development of innovative modulation strategies [20]–[36]. Although new topologies, such as multi-level or 2-level topologies with additional switches, increase the cost and require complex control strategies, innovative modulation techniques offer the advantage of streamlined integration, requiring minimal additional space and noncomplex modifications. Therefore, this study focuses on modulation-based solutions.

In [32], a modified space vector modulation (SVM) was proposed, which decreases the amount and amplitude of common-mode voltage by reducing the number of transitions to six. In [33], the switching patterns were updated to reduce the CMV for two PWM algorithms: near-state PWM and the modified form of the active zero-state PWM. Here, the proposed method performs well for both low and high modulation indices. In [34], a CMV reduction method was developed for carrier-based sinusoidal PWMs (CB-SPWM) where carrier phase shifts between the legs of the 3ph-VSI (0° , 120° , and 240°) are introduced. In [35], the CMV reduction for CB-SPWM was improved by introducing the sampling-time-adaptive carrier-phase-shift method, which yields carrier phase shifts of 0° or 180° with respect to the reference voltage of the legs. The methods discussed above reduce the amount and magnitude of CMV, but they cannot eliminate the generated dv/dt . Other modulation methods to eliminate dv/dt have been proposed in a dual-inverter structure for 3 phases or a single-inverter structure for multiphase inverters [36]. In these modulation methods, the main idea is to synchronize the pulse patterns of the inverter legs such that they cancel the rising and falling dv/dt of the legs. However, this synchronization procedure is infeasible when using conventional sinusoidal SPWM or SVM in a single inverter with a 2L-3ph system. This study, therefore, introduces a new modulation technique based on CB-SPWM to make 2L-3ph motor drives feasible for dv/dt cancellation based on PWM synchronization. A third-harmonic square wave is injected in the proposed modulation

technique, and the variable carrier phase shifts that are updated for each switching period are introduced. In this way, complete dv/dt cancellation is achieved in an ideal case. The main contributions of the study are characterized by:

- 1) A new modulation method based on conventional SPWM is introduced to reduce CMV, allowing it to work alongside traditional motor drives.
- 2) This modulation method can be easily implemented with a minor additional computational burden, making it suitable for use with off-the-shelf microcontrollers or FPGAs.
- 3) Although this method is designed for 2-level, 3-phase drive systems, it is not limited to them. It can be easily extended to multilevel, multiphase drives, making it a versatile solution.
- 4) By eliminating the generated common-mode dv/dt , this method increases the lifetime of motor insulations and bearings.

The rest of this paper is organized as follows. Section II presents the system structure and analyzes the common-mode voltage in the 2L-3Ph-VSIs. Section III presents the mathematical background of the mitigation method and explains the injection and adaptive carrier phase shifts. Section IV presents the experimental setup and validation. Section V discusses practical issues in the implementation and provides a generic form for multiphase machines. Finally, Section VI concludes the study.

II. SYSTEM STRUCTURE AND COMMON MODE VOLTAGE ANALYSIS

A. System Structure

Fig. 1 shows a traditional 2L-3Ph motor drive system, and a simplified common mode equivalent circuit diagram. These equivalent circuit parameters represent the stray capacitances between the windings and the cores of the electrical motor.

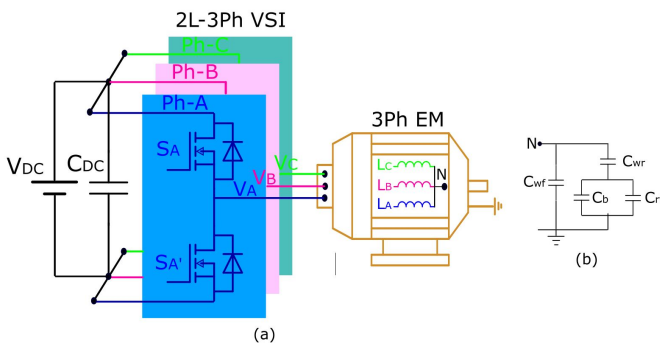


Fig. 1. a) The system structure of a conventional 2-level 3-phase motor drive system. b) Equivalent common mode circuit diagram.

C_{wf} , C_{wr} , C_{rf} , and C_b represent the stray capacitances between the stator winding and frame, stator winding and rotor, rotor and frame, and bearing and frame, respectively. These stray capacitances play a significant role in common-mode currents caused by the generated dv/dt . Their influence becomes more noticeable at high frequencies because they provide low-impedance paths for the common-mode currents

to flow through. Therefore, if the motor-side parameters remain unchanged, reducing the high-frequency common-mode voltage at the inverter output may be the only solution to prevent failures caused by these common-mode currents.

B. Common Mode Voltage Analysis

Regarding the system shown in Fig. 1, the CMV is calculated by taking the average value of the three-phase voltages as given by

$$V_{CM} = \frac{1}{3}(V_A + V_B + V_C), \quad (1)$$

where V_A , V_B , and V_C are phase-to-neutral voltages, altered between $-V_{DC}/2$ and $+V_{DC}/2$, where V_{DC} is the pole-to-pole input voltage. Then, the inverter is driven by carrier-based (CB) SPWM that can be produced by comparing sinusoidal references with triangular carrier signals. Conventionally, it is prevalent to use a common single-carrier signal for all phases with 120° cyclically phase-shifted sinusoidal reference signals for each phase. In such a case, the carrier, reference signals, and switching pulses during a fundamental period are shown in Fig. 2.a and Fig. 2.b.

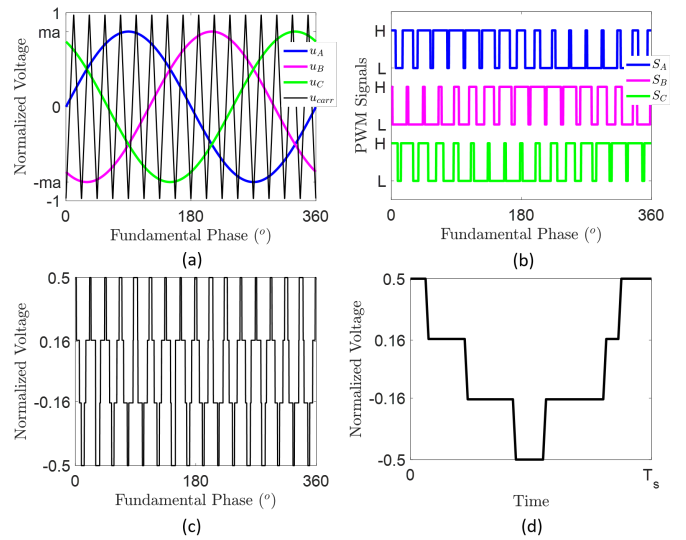


Fig. 2. The key waveforms of 2L-3Ph VSI. a) Reference and carrier signals of conventional SPWM. b) PWM signals for the high-side switch of each leg. c) Normalized CMV of the 2L-3Ph VSI driven by SPWM. d) Close-view of the normalized CMV for a single switching period.

Therefore, the CMV, which is calculated using (1) along with the fundamental and switching periods, is shown in Fig. 2.c and Fig. 2.d, respectively. It is observed that the common-mode voltage (CMV) frequency corresponds to the switching frequency and its multiples, with switching transitions (dv/dt) occurring six times within a switching period. In addition, the CMV voltages cause current to flow through the stray capacitances, as represented in the common-mode equivalent circuit. Therefore, the impact of the CMV increases with both the switching frequency and switching speed of the MOSFETs. However, selecting the switching frequency and speed requires a holistic approach to optimize the overall system. For instance, the switching frequency is chosen based

on the motor's requirements for differential current ripple and harmonic losses, whereas the switching speed is selected based on the inverter's electrical loss and thermal constraints. Consequently, utilizing new modulation techniques could potentially mitigate CMV without altering the switching frequency and speed. This approach can significantly improve the motor drive system performance without introducing additional cost and complexity.

III. THE PROPOSED MITIGATION METHODOLGY

The proposed method investigates a single switching period, instead of a fundamental period. If CMV is eliminated within a switching period at each operating point of the SPWM, the overall CMV is eliminated. Fig. 3 illustrates carrier and reference signals, phase-to-neutral output voltages, and CMV during a single switching period for both a single carrier signal and multiple carrier signals with phase shifts, represented as $\phi_{cA} = 0$, $\phi_{cB} = \frac{t_B}{T_s} 2\pi$, $\phi_{cC} = \frac{t_C}{T_s} 2\pi$ where t_A is taken as a reference and is selected as zero; t_B , and t_C are time delays between the carrier signals.

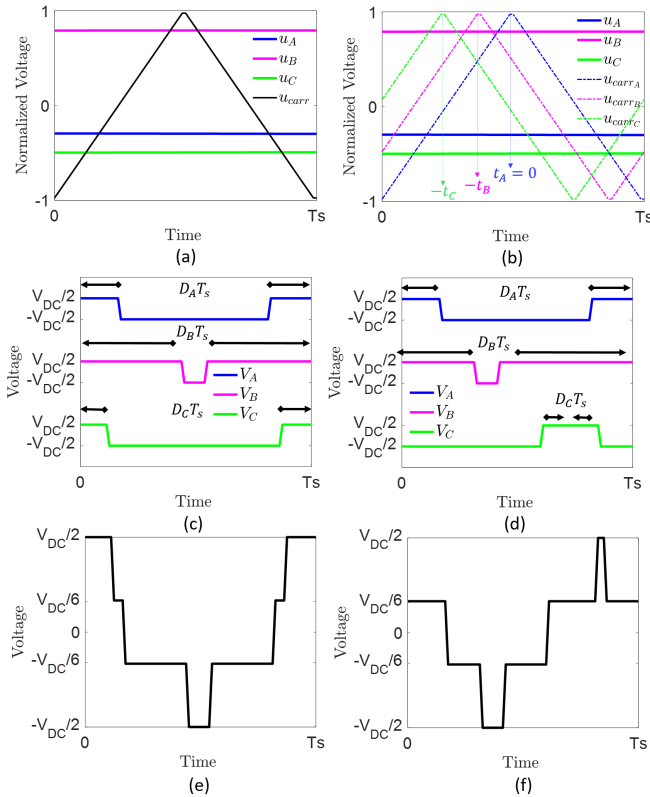


Fig. 3. Closeup waveforms for a single switching period. Carrier and reference voltages: a) Single-carrier SPWM. b) Multi-carrier SPWM. Phase-to-neutral output voltages: c) Single-carrier SPWM. d) Multi-carrier SPWM. Common mode voltages: e) Single-carrier SPWM. f) Multi-carrier SPWM.

Although the duty ratios are given the same for both single-carrier and multiple-carrier SPWM, Fig. 3.e and 3.f show that the shapes of CMVs are different, so the key idea to eliminate CMV is to introduce multiple carriers with distinct phase shifts. These duty ratios within switching periods are

determined by the sinusoidal reference signals, which are given by

$$\begin{aligned} u_A &= m_a \sin(2\pi f_o t), \\ u_B &= m_a \sin(2\pi f_o t - \frac{2\pi}{3}), \\ u_C &= m_a \sin(2\pi f_o t + \frac{2\pi}{3}), \end{aligned} \quad (2)$$

where m_a is the modulation index, f_o is the fundamental frequency, and t is time. Accordingly, the duty ratios can be calculated by

$$\begin{aligned} D_A &= \frac{1 + u_A}{2}, \\ D_B &= \frac{1 + u_B}{2}, \\ D_C &= \frac{1 + u_C}{2}. \end{aligned} \quad (3)$$

Then, within a switching period, the phase voltages can be approximated as constant, following the duty ratios. Hence, the output voltages within a switching period (denoted as V_h) can be represented by a Fourier series, as given by

$$V_h = \sum_{n=-\infty}^{\infty} c_n e^{j2\pi n f_{sw} t}, \quad (4)$$

where f_{sw} is the switching frequency, and c_n are the Fourier coefficients, which can be calculated with respect to the harmonic number as given by

$$c_n = \begin{cases} \frac{V_{DC}}{-j2\pi n} (e^{-jn(2\pi D + \phi_C)} - e^{-jn\phi_C}) & \text{if } n \neq 0, \\ \frac{V_{DC}}{2} D & \text{if } n = 0, \end{cases} \quad (5)$$

where D are duty ratios and ϕ_C are carrier phase shifts. In fine, to eliminate overall high-frequency CMV, the summation of harmonic components of the phases, and so the summation of c_n s, should be zero, as given by

$$\frac{1}{3} \sum_{ph}^{(A,B,C)} \sum_{n=1}^{\infty} c_{n_{ph}} e^{j2\pi n f_{sw} t} = 0. \quad (6)$$

In this sum, the n^{th} harmonic Fourier coefficients for CMV are calculated by

$$c_{n_{CMV}} = c_{n_A} + c_{n_B} + c_{n_C}. \quad (7)$$

The aim is to find carrier phase shift conditions that give $c_{n_{CMV}}$ zero in this sum concerning duty ratios that follow the reference signals. To find the conditions, the Fourier series of $c_{n_{CMV}}$ can be explicitly written as

$$\begin{aligned} c_{n_{CMV}} &= \\ \frac{V_{dc}}{-j2\pi n} &\left(e^{-jn(2\pi D_A + \phi_{cA})} + e^{-jn(2\pi D_B + \phi_{cB})} + e^{-jn(2\pi D_C + \phi_{cC})} \right. \\ &\quad \left. + e^{-jn\phi_{cA}} + e^{-jn\phi_{cB}} + e^{-jn\phi_{cC}} \right). \end{aligned} \quad (8)$$

To ensure the equation holds for all harmonic orders, the phasors, six exponential terms in (8), can be organized into symmetric sets, such as triangular or hexagonal configurations, to enable overall cancellation. However, since the phasors

depend on the harmonic number (n), a solution tailored for one harmonic may not generalize to all harmonic numbers. Another approach is pairwise cancellation, where each duty-modulated term is matched with a corresponding carrier-phase term, ensuring the summation cancels out for each harmonic. This avoids harmonic-specific constraints but does not yield a unique solution, as different pairings produce equally valid results. For practical purposes, one such solution can be selected, leading to the following system of equations:

$$\begin{aligned} n(2\pi D_A + \phi_{cA}) &= n\phi_{cB}, \\ n(2\pi D_B + \phi_{cB}) &= n\phi_{cC}, \\ n(2\pi D_C + \phi_{cC}) &= n\phi_{cA}. \end{aligned} \quad (9)$$

Then, by substituting the second and third equations of (9) into the first, the single equation is achieved as given by

$$2\pi D_A + \phi_{cA} + 2\pi D_B + \phi_{cB} + 2\pi D_C + \phi_{cC} = \phi_{cB} + \phi_{cC} + \phi_{cA}. \quad (10)$$

Therefore, vanishing all harmonic terms depends on the condition given by

$$2\pi(D_A + D_B + D_C) \equiv 0 \pmod{2\pi}. \quad (11)$$

Accordingly, the sum of the duty ratios should be an integer (1,2,3, etc). When the sum of duty cycles meets the integer

condition, the required carrier phase shifts can be calculated regarding the duty ratios as given by

$$\begin{aligned} \phi_{cA} &= 0, \\ \phi_{cB} &= (D_A + D_B)\pi, \\ \phi_{cC} &= (D_A + 2D_B + D_C)\pi, \end{aligned} \quad (12)$$

where, for simplicity, assume that $\phi_{cA} = 0$. An exemplary demonstration of two distinct cases (the sum of duty cycles is 1 and the sum of duty cycles is 2) can be seen in Fig. 4. However, in conventional SPWM, the sum of the duty ratios is constant ($3/2$) for each operating point, which is given by

$$D_{sum} = \frac{1+u_A}{2} + \frac{1+u_B}{2} + \frac{1+u_C}{2} = \frac{3}{2} \quad (13)$$

since the reference signals (u_A, u_B, u_C) create a positive sequence.

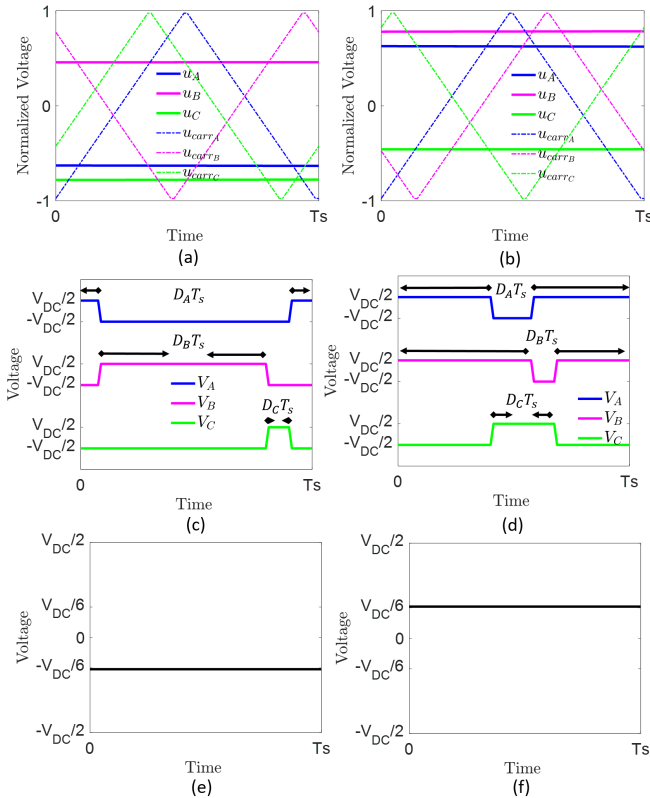


Fig. 4. Closeup waveforms for a single switching period. Carrier and reference voltages: a) $D_{sum} = 1$, b) $D_{sum} = 2$. Phase-to-neutral output voltages: c) $D_{sum} = 1$, d) $D_{sum} = 2$. Common mode voltages: e) $D_{sum} = 1$, f) $D_{sum} = 2$.

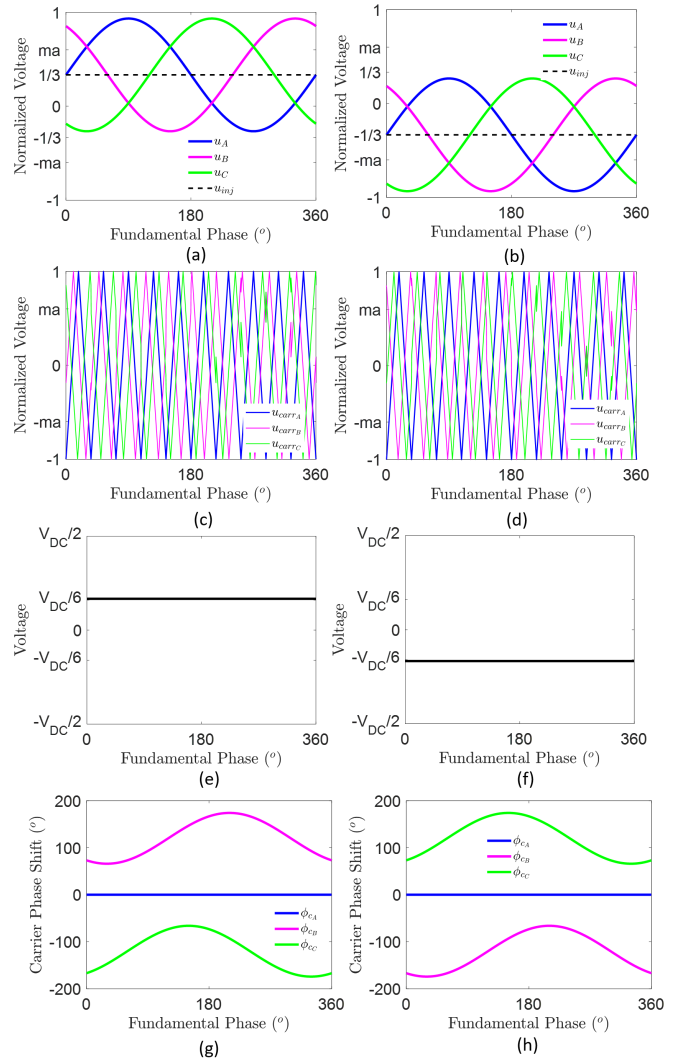


Fig. 5. Example DC injections with $m_a = 0.6$. Reference signals. a) $+1/3$ DC Injection. b) $-1/3$ DC Injection. Carrier signals. c) $+1/3$ DC Injection. d) $-1/3$ DC Injection. CMV. e) $+1/3$ DC Injection. f) $-1/3$ DC Injection. Carrier-phase-shift values. g) $+1/3$ DC Injection. h) $-1/3$ DC Injection.

Hence, the reference voltages should be updated to achieve the sum of the integer number. The first idea is to inject

a DC signal (as a common mode) into the reference signal to hold the integer sum condition. Each reference signal can be subtracted or added by $1/3$ to achieve the sum of duty cycles of 1 or 2. Fig. 5.a and 5.b show the positive and negative DC injection to the reference signals. Additionally, the required carrier phase shifts for DC-injected SPWM to cancel the CMV are illustrated in Fig. 5.g and 5.h. However, in this situation, the maximum modulation index is limited to $2/3$ because of $1/3$ DC injection. The modulation index limitation should be enhanced to increase DC-link voltage utilization. Therefore, the second idea is to inject a third harmonic square wave that is like altering positive and negative DC injections periodically. Although the amplitude of the square wave is the same with DC injection ($1/3$), the maximum modulation index is augmented. The reference signals and injected square wave are illustrated in Fig 6.a and 6.b.

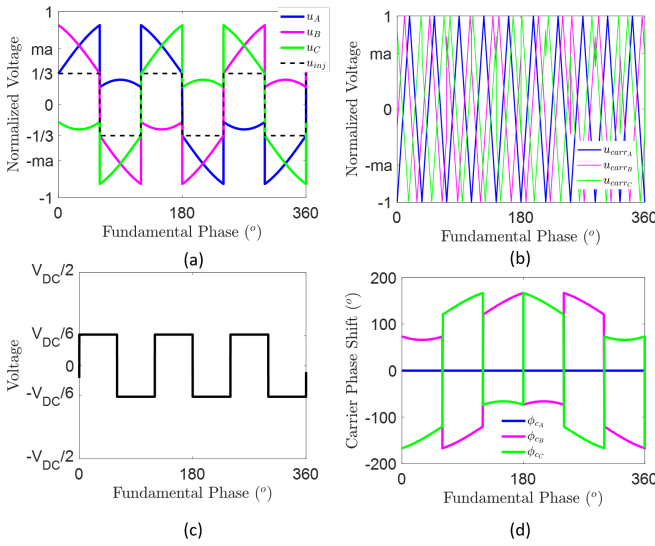


Fig. 6. Example square wave injection with $m_a = 0.6$. a) Reference signals. b) Carrier signals. c) CMV. d) Carrier-phase-shift values.

As seen in Fig. 6.a, the maximum modulation index can be calculated at the 60° of phase-A. The maximum value of the reference signal at this angle should be 1, and so the maximum modulation index is found to be $\frac{4}{3\sqrt{3}}$ as calculated by

$$m_a \sin\left(\frac{\pi}{3}\right) + \left(\frac{1}{3}\right) = 1, \quad (14)$$

$$m_a = \frac{1 - (\frac{1}{3})}{\sin(\frac{\pi}{3})} = \frac{4}{3\sqrt{3}} \approx 0.77.$$

This intentional signal injection introduces low-frequency common-mode voltage components, but as discussed before, the common mode currents become dominant at high frequencies. Also, low-frequency common-mode voltage excitation is a common practice in motor drives to increase DC-link utilization. Therefore, the effect of the third-harmonic square wave injection becomes very low compared to high-frequency CMV. Consequently, the overall modulation strategy is shown in Algorithm 1.

IV. EXPERIMENTAL VALIDATION

An experimental setup, as shown in Fig. 7, has been established to validate the proposed method. The setup comprises Imperix SiC power modules (PEB8024 half-bridges) to form

Algorithm 1 Proposed Modulation Technique

Input: The main controller generates sinusoidal references:

$$u_{(A,B,C)} = m_a \sin(2\pi f_o t - \theta_{o(A,B,C)})$$

Step 1: Generating square-wave signal:

$$u_{inj} = \frac{1}{3} \text{sgn}[-\sin(2\pi f_o t - \theta_{oA})]$$

Step 2: Summing the sinusoidal references and square-wave signal:

$$u_{(A^*,B^*,C^*)} = u_{(A,B,C)} + u_{inj}$$

Step 3: Calculating updated duty ratios (D_A^* , D_B^* , D_C^*):

$$D_{(A,B,C)}^* = \frac{1 + u_{(A^*,B^*,C^*)}}{2}$$

Step 4: Calculating carrier phase shifts:

$$\begin{aligned} \phi_{cA} &= 0 \\ \phi_{cB} &= (D_A^* + D_B^*)\pi \\ \phi_{cC} &= (D_A^* + 2D_B^* + D_C^*)\pi \end{aligned}$$

Step 5: Generating carrier signals:

$$u_{carr(A,B,C)} = \text{tri}(t, \phi_{c(A,B,C)})$$

Step 6: Comparing the carrier signals with the duty ratios:

$$S_{(A,B,C)} = \begin{cases} 1, & \text{if } D_{(A,B,C)}^* \geq u_{carr(A,B,C)}(t) \\ 0, & \text{otherwise} \end{cases}$$

Output: Generating PWM signals (S_A , S_B , S_C)

a 2-Level 3-Phase inverter. To implement the proposed modulation strategy, the Imperix B-Box RCP controller is utilized. The inverter system is then connected to an induction machine to evaluate the proposed method under loaded conditions.

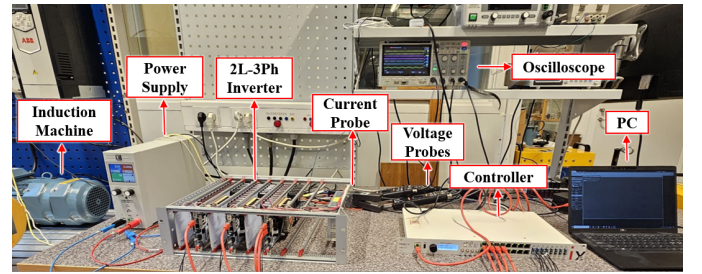


Fig. 7. Experimental setup of 2-level 3-phase inverter system.

A. No-load Test

The phase voltages were measured under no-load conditions to evaluate dv/dt reduction performance. Fig. 8 shows the common-mode voltage (CMV) waveforms with and without the proposed mitigation method. To emphasize the differences, a close-up analysis was also performed at one operating point, focusing on two switching periods, which is shown in Fig. 9. It was observed that the proposed method effectively reduces the high-frequency common-mode voltage components while introducing a low-frequency (third-harmonic

fundamental) square-wave common-mode voltage. However, a slightly high-frequency common-mode voltage still exists due to oscillations in the phase voltages caused by layout parasitics and measurement non-idealities in the experimental setup.

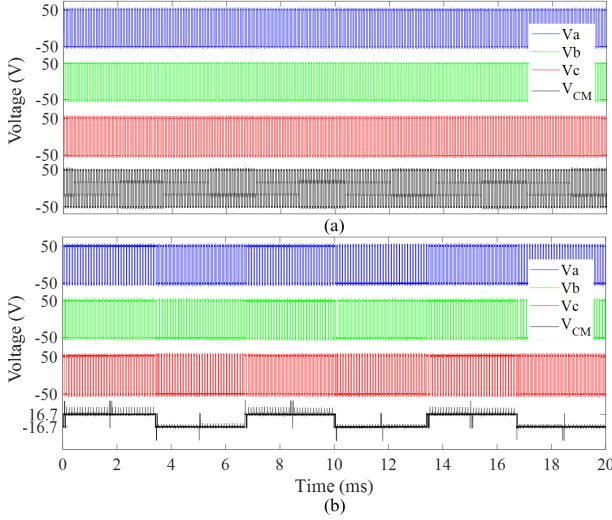


Fig. 8. The phase voltages (V_A, V_B , and V_C) of the inverter and common-mode voltage (V_{CM}). a) Conventional SPWM. b) The proposed method.

Furthermore, the system's performance across different operating conditions was analyzed. The corresponding duty ratios and the calculated carrier phase shifts for the selected operating conditions are provided in Table I. The phase voltage waveforms are shown in Fig. 10. The results proved that successful pulse synchronization was achieved by the proposed method. Although high-frequency oscillations exist in the measurements, the primary dv/dt transitions remain appropriately aligned. This alignment confirms the robustness of the proposed method against practical challenges.

B. Load Test

The dynamic behavior of switching transitions varies with phase currents, requiring experimental validation under loaded conditions. The system was tested with the following parameters: modulation index ($m_a = 0.6$), output frequency ($f_o = 20$ Hz), switching frequency ($f_{sw} = 10$ kHz), and DC-link voltage ($V_{DC} = 100$ V). Fig. 11 shows the phase

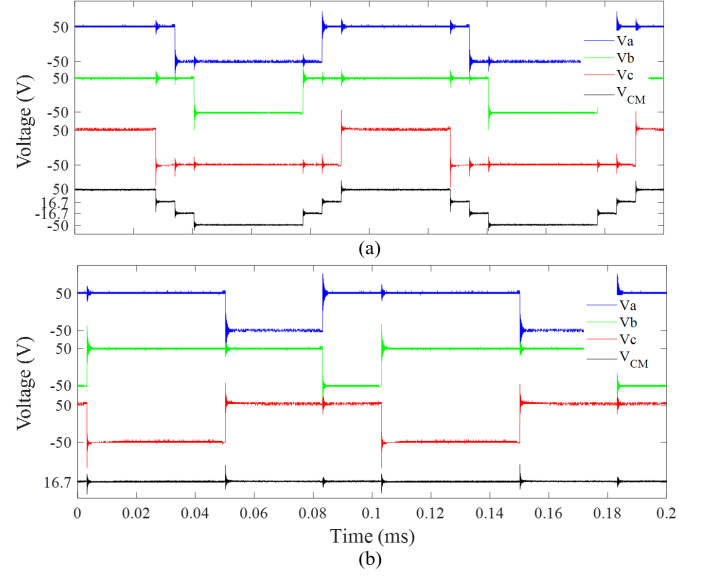


Fig. 9. The phase voltages (V_A, V_B , and V_C) of the inverter and common-mode voltage (V_{CM}). a) Conventional SPWM. b) The proposed method.

voltages and phase current, including a zoomed-in view of a random switching event. It was observed that the dv/dt of the turn-on and turn-off transitions are different because of the phase currents. These differences result in partial dv/dt elimination rather than full cancellation. Solutions for achieving full cancellation under all operating conditions will be discussed in the next section.

V. DISCUSSION

A. Modulation Limitations

In the proposed method, the modulation index is limited to $\frac{4}{3\sqrt{3}} \approx 0.77$. This limitation affects performance parameters like efficiency and power density. However, the use of a low modulation index might be met in specialized applications like aerospace, where high-voltage DC-link systems are utilized, and lower AC output voltages are demanded because of machine design aspects [37], [38]. Although the modulation index is considered a critical design parameter, it is possible to achieve equivalent motor performance with a lower

TABLE I
DUTY RATIOS & PHASE SHIFTS FOR DISTINCT OPERATING POINTS

	Fundamental Phase Angle ($^\circ$) at $m_a = 0.3$				Fundamental Phase Angle ($^\circ$) at $m_a = 0.6$			
	0	90	180	270	45	135	225	315
D_A	0.6500	0.5000	0.3500	0.5000	0.7121	0.2879	0.2879	0.7121
D_B	0.4250	0.6299	0.5750	0.3701	0.5776	0.7898	0.4224	0.2102
D_C	0.4250	0.3701	0.5750	0.6299	0.2102	0.4224	0.7898	0.5776
$\sum D$	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000
D_A^*	0.4833	0.6667	0.5167	0.3333	0.8788	0.1212	0.1212	0.8788
D_B^*	0.2583	0.7966	0.7417	0.2034	0.7443	0.6231	0.2557	0.3769
D_C^*	0.2583	0.5368	0.7417	0.4632	0.3769	0.2557	0.6231	0.7443
$\sum D^*$	1.0000	2.0000	2.0000	1.0000	2.0000	1.0000	1.0000	2.0000
ϕ_{cA}	0	0	0	0	0	0	0	0
ϕ_{cB}	133.5000	263.3827	226.5000	96.6173	292.1600	133.9762	67.8400	226.0238
ϕ_{cC}	226.5000	143.3827	133.5000	216.6173	133.9762	292.1600	226.0238	67.8400

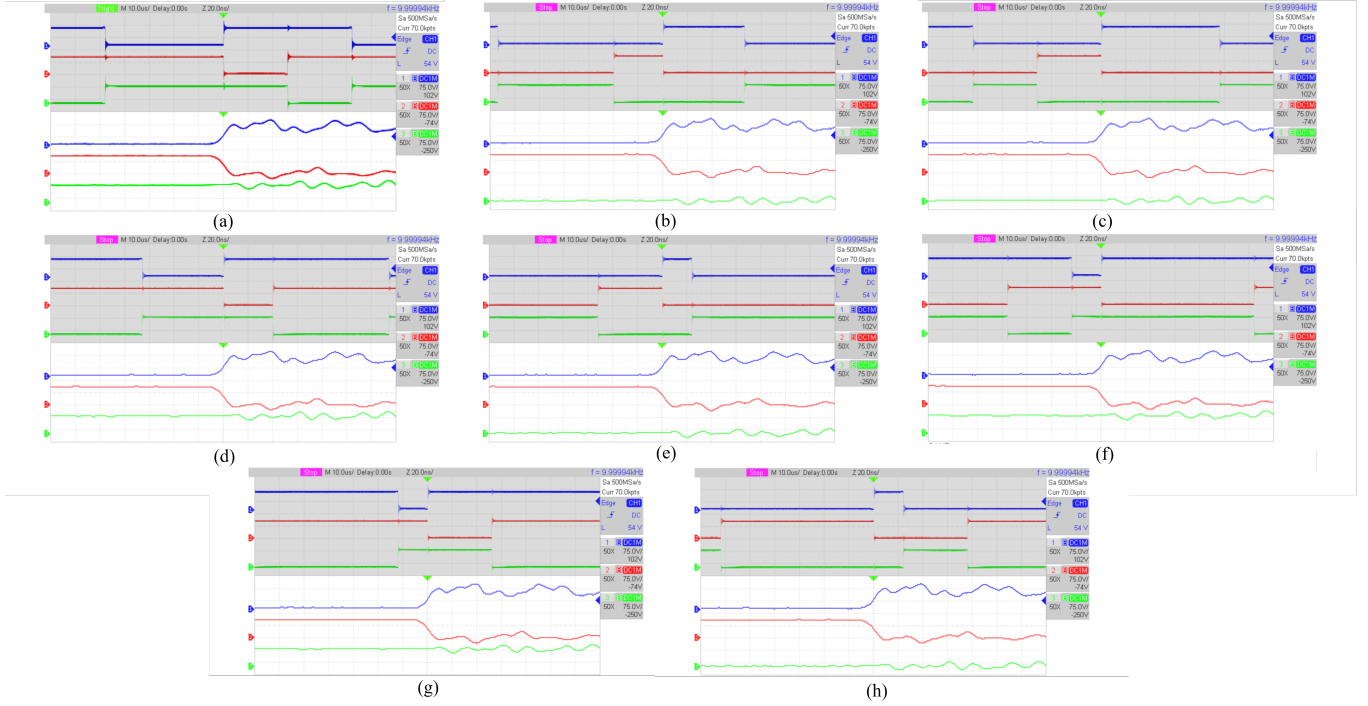


Fig. 10. The phase voltages (V_A, V_B , and V_C) of the inverter. Blue color : V_A , Green color : V_B , Red color : V_C . a) $m_a = 0.3$ Fundamental phase: 0° , b) $m_a = 0.3$ Fundamental phase: 90° , c) $m_a = 0.3$ Fundamental phase: 180° , d) $m_a = 0.3$ Fundamental phase: 270° , e) $m_a = 0.6$ Fundamental phase: 45° , f) $m_a = 0.6$ Fundamental phase: 135° , g) $m_a = 0.6$ Fundamental phase: 225° , h) $m_a = 0.6$ Fundamental phase: 315° .

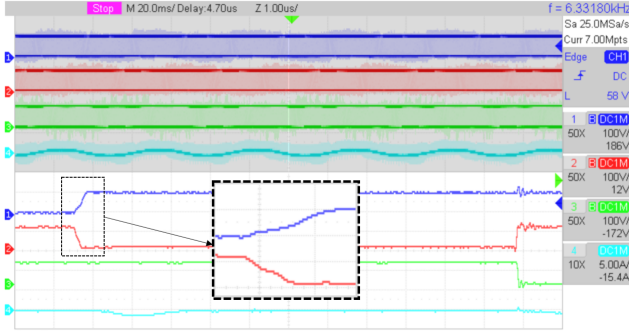


Fig. 11. The phase voltages (V_A, V_B , and V_C) of the inverter, common mode voltage (V_{CM}) and phase-A current (I_{phA}). Blue color : V_A , Green color : V_B , Red color : V_C , Turquoise color: I_{phA} .

modulation index at the same DC-link voltage by configuring motor windings in either series or parallel.

B. Impact of Non-idealities

First, the dv/dt depends on the phase currents, which means that the rising and falling edges of different legs might not compensate for each other. Even without precautions, a natural partial reduction in dv/dt still occurs. A more effective solution to solve dv/dt variations is to introduce variable gate resistance, which allows a constant dv/dt under all operating conditions [39]. Second, the impact of dead time and gate driver delay mismatches, which change the phase shift calculations, must be accounted for. To overcome this problem, the gate driver delays can be measured, and the dead times appropriately adjusted, incorporating these corrections into the phase calculations.

C. Generalization for Multi-phase Inverters

As derived in (11), the sum of duty ratios should be an integer to be able to synchronize PWMs and cancel the generated dv/dt . Even if the number of phases is increased, this condition does not change, and the sum of duty ratios for multi-phase inverters using SPWM can be calculated by

$$D_{sum} = \frac{\#Phases}{2}. \quad (15)$$

First, for the even numbers of phases, the sum of the duty ratio becomes an integer intrinsically, which means that there is no need for square wave injection. Therefore, the limitation of modulation disappears inherently. Second, for odd numbers of phases, square wave injection is still necessary, but the amplitude of the injection becomes smaller while increasing the phase number, which means that the modulation limitation decreases. Consequently, the synchronization method can be easily employed in multi-phase inverters, and increasing the phase number makes the drawback of modulation limitation lower.

VI. CONCLUSION

This paper presents a novel PWM method to effectively reduce common-mode voltage caused by the generated dv/dt in 2-level 3-phase voltage source inverters. By synchronizing the PWM signals of the inverter legs, facilitated through third harmonic square wave injection and adaptive carrier phase shift, the proposed method mitigates the CMV issue. In this study, a comprehensive analysis of the modulation by the Fourier series was given, and analytical calculations were conducted to calculate the required phase shifts and injected signals. Then, the proposed technique's practicality and robustness were thoroughly validated through experimental tests

with SiC devices. The experimental results confirm that this method offers a promising approach for minimizing CMV in high-performance inverter systems without increasing system cost or complexity.

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